THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The cost of running a 1.5 kW heater for 4 hours at Rs 6 per unit is 1

(a) Rs 60 **(b)** Rs 24 **(c)** Rs 9 **(d)** Rs 36

2. A wire of radius r has resistance R. A wire of the same material and length but radius 2r has a resistance 1

(a) 2R **(b)** R/2 **(c)** R/4 **(d)** 4R

3. Two identical bulbs connected in parallel to a battery, compared to the series connection, are 1

(a) equally bright **(b)** dimmer **(c)** off **(d)** brighter

4. Which of the following lamps has the highest resistance (all rated for 220 V)? 1

(a) 25 W **(b)** 60 W **(c)** 100 W **(d)** 40 W

5. A wire has resistance 12 ohm. It is cut into three equal pieces. The resistance of each piece is 1

(a) 12 ohm **(b)** 1/4 ohm **(c)** 4 ohm **(d)** 36 ohm

6. The current through the 4 ohm resistor when 4 ohm and 12 ohm are in parallel across 12 V is 1

(a) 12 A **(b)** 1 A **(c)** 4 A **(d)** 3 A

7. A 60 W bulb is used for 10 hours. The energy consumed in joules is 1

(a) 6×10^4 J **(b)** 2.16×10^6 J **(c)** 600 J **(d)** 3.6×10^6 J

8. The resistivity of a material depends on 1

(a) its area **(b)** the current through it

(c) its length **(d)** its nature and temperature

9. The total current drawn from a 12 V battery by a 4 ohm and a 12 ohm resistor in parallel is 1

(a) 1 A **(b)** 4 A **(c)** 3 A **(d)** 0.75 A

10. Which of these energy conversions occurs in an electric heater? 1

(a) Electrical to heat **(b)** Light to heat

(c) Heat to electrical **(d)** Chemical to electrical

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A bulb is marked 220 V, 55 W. Find the current. 2

12. How will the resistance of a wire change if its diameter is halved? 2

13. Why does a heater coil get hot while the wires connecting it remain cool? 2

14. A potential difference of 20 V is applied across a 5 ohm and a 15 ohm resistor in series. Find the voltage across the 15 ohm resistor. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A conductor of resistance 5 ohm has current 3 A for 2 minutes. Find the heat produced. 3

16. The current in a conductor is 2 A at 8 V. What current flows at 20 V if the temperature does not change? 3

17. An electric kettle draws 5 A at 220 V. How long does it take to produce 3.3×10^5 J of heat? 3

OR

A bulb is rated 40 W at 220 V. Find the resistance and the current. If it is used at 110 V, find the power consumed approximately. 3

18. A student plots V versus I for two resistors and finds the slopes are 2 ohm and 6 ohm. Which resistor is thicker if they are the same length and material? 3

OR

A resistor of 12 ohm is connected to a battery of unknown voltage and carries 2 A. Find the voltage and the power. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A 1.5 kW geyser and a 100 W fan, a 60 W bulb are used for 4 hours, 8 hours and 6 hours respectively every day. Calculate the monthly (30 days) bill at Rs 7 per unit. 5

OR

In a circuit, two bulbs of 40 W and 60 W (each rated 220 V) are connected 5
(a) in series
(b) in parallel across 220 V. Which glows brighter in each case? Explain.

20. Two lamps, one rated 100 W at 220 V and the other 25 W at 220 V, are connected in series across 220 V. Which glows brighter? Find the current. 5

OR

An electric iron of 1 kW is used on 220 V for 2 hours daily. 5
(a) Find the current drawn
(b) the resistance of its element
(c) the energy used in a month of 30 days in kWh and in joules.

END OF QUESTION PAPER Total: 40 Marks

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